

Supplementary File S1

Trustworthy Medication Recommendation: A Safety-Centered Survey of Knowledge-Grounded and LLM-Assisted Methods

Source-Specific Search Strategies and Study-Identification Details

Repository: https://github.com/Zafar-southeast/Medication_Recommendation_Survey

This supplementary file documents the source-specific searches, venue-specific coverage verification, PRISMA identification counts, and the studies included in the qualitative synthesis.

S1. Search Strategy Overview

The systematic literature search comprised a preliminary scoping phase followed by a formal systematic search, supplemented by independent venue-specific coverage verification.

Preliminary Scoping Search: Broad task-specific terms were used to assess the approximate volume of literature and inform refinement of the formal search.

Formal Systematic Search: The broad terms were combined with modality-, architecture-, and trustworthiness-related terms to define the formal corpus.

Venue-Specific Coverage Verification: Selected major journals and conference proceedings were independently screened as a coverage check. No additional unique records were identified beyond those already captured in the master source-based set.

Preliminary scoping terms: "medication recommendation", "drug recommendation", "prescription prediction".

Data modalities: EHR, electronic health record, clinical narratives, multimodal fusion, temporal dynamics.

Architectural terms: drug-drug interactions, knowledge graphs, ontologies, hypergraphs, Transformers, GNNs, large language models, LLMs, retrieval-augmented generation, RAG.

Trustworthiness/evaluation terms: hallucination, calibration, robustness, interpretability, generalization.

All searches and screenings were executed or completed by 20 May 2026.

S2. Source-Specific Search Strategies and Retrieved Records

The preliminary scoping-search counts are reported for transparency but were not included in the PRISMA flow. The formal systematic searches yielded 132 records before duplicate removal and formed the basis of the PRISMA identification stage.

Table S1. Source-specific search strategies, execution dates, applied limits, and records retrieved.

Source	Search stage	Final search date	Exact executed query	Limits applied	Records retrieved
PubMed	Preliminary Scoping	20 May 2026	("medication recommendation"[TIAB] OR "drug recommendation"[TIAB] OR "prescription prediction"[TIAB])	Title/Abstract; English; 1 Jan 2022-20 May 2026	367
PubMed	Formal Systematic	20 May 2026	("medication recommendation"[TIAB] OR "drug recommendation"[TIAB] OR "prescription prediction"[TIAB]) AND ("EHR"[TIAB] OR "electronic health record"[TIAB] OR "clinical narratives"[TIAB] OR "multimodal fusion"[TIAB] OR "temporal dynamics"[TIAB] OR "drug-drug interactions"[TIAB] OR "knowledge graphs"[TIAB] OR "ontologies"[TIAB] OR "hypergraphs"[TIAB] OR "Transformers"[TIAB] OR "GNNs"[TIAB] OR "large language models"[TIAB] OR "LLMs"[TIAB])	Title/Abstract; English; 1 Jan 2022-20 May 2026	24

Source	Search stage	Final search date	Exact executed query	Limits applied	Records retrieved
			OR "retrieval-augmented generation"[TIAB] OR RAG[TIAB] OR hallucination[TIAB] OR calibration[TIAB] OR robustness[TIAB] OR interpretability[TIAB] OR generalization[TIAB])		
IEEE Xplore	Preliminary Scoping	20 May 2026	("Document Title":"medication recommendation" OR "Document Title":"drug recommendation" OR "Document Title":"prescription prediction")	Document Title; 1 Jan 2022-20 May 2026	211
IEEE Xplore	Formal Systematic	20 May 2026	("Document Title":"medication recommendation" OR "Document Title":"drug recommendation" OR "Document Title":"prescription prediction") AND ("Abstract":"EHR" OR "Abstract":"electronic health record" OR "Abstract":"clinical narratives" OR "Abstract":"multimodal fusion" OR "Abstract":"temporal dynamics" OR "Abstract":"drug-drug interactions" OR "Abstract":"knowledge graphs" OR "Abstract":"ontologies" OR "Abstract":"hypergraphs" OR "Abstract":"Transformers" OR "Abstract":"GNNs" OR "Abstract":"large language models" OR "Abstract":"LLMs" OR "Abstract":"retrieval-augmented generation" OR "Abstract":"RAG" OR "Abstract":"hallucination" OR "Abstract":"calibration" OR "Abstract":"robustness" OR "Abstract":"interpretability" OR "Abstract":"generalization")	Document Title and Abstract; 1 Jan 2022-20 May 2026	18
ACM Digital Library	Preliminary Scoping	20 May 2026	((("medication recommendation") OR ("drug recommendation")) OR ("prescription prediction"))	Full-text; 1 Jan 2022-20 May 2026	145
ACM Digital Library	Formal Systematic	20 May 2026	((("medication recommendation" OR "drug recommendation" OR "prescription prediction")) AND ((EHR OR "electronic health record" OR "clinical narratives" OR "multimodal fusion" OR "temporal dynamics" OR "drug-drug interactions" OR "knowledge graphs" OR "ontologies" OR "hypergraphs" OR "Transformers" OR "GNNs" OR "large language models" OR "LLMs" OR "retrieval-augmented generation" OR "RAG" OR "hallucination" OR "calibration" OR "robustness" OR "interpretability" OR "generalization")))	Full-text; 1 Jan 2022-20 May 2026	20
Scopus	Preliminary Scoping	20 May 2026	TITLE-ABS-KEY("medication recommendation" OR "drug	Title/Abstract/Keywords; English; 1 Jan 2022-20 May 2026	402

Source	Search stage	Final search date	Exact executed query	Limits applied	Records retrieved
			recommendation" OR "prescription prediction")		
Scopus	Formal Systematic	20 May 2026	TITLE-ABS-KEY(("medication recommendation" OR "drug recommendation" OR "prescription prediction") AND ("EHR" OR "electronic health record" OR "clinical narratives" OR "multimodal fusion" OR "temporal dynamics" OR "drug-drug interactions" OR "knowledge graphs" OR ontologies OR hypergraphs OR Transformers OR GNNs OR "large language models" OR LLMs OR "retrieval-augmented generation" OR RAG OR hallucination OR calibration OR robustness OR interpretability OR generalization))	Title/Abstract/Keywords; English; 1 Jan 2022-20 May 2026	27
Web of Science	Preliminary Scoping	20 May 2026	TS=("medication recommendation" OR "drug recommendation" OR "prescription prediction")	Topic; English; 1 Jan 2022-20 May 2026	314
Web of Science	Formal Systematic	20 May 2026	TS(("medication recommendation" OR "drug recommendation" OR "prescription prediction") AND ("EHR" OR "electronic health record" OR "clinical narratives" OR "multimodal fusion" OR "temporal dynamics" OR "drug-drug interactions" OR "knowledge graphs" OR ontologies OR hypergraphs OR Transformers OR GNNs OR "large language models" OR LLMs OR "retrieval-augmented generation" OR RAG OR hallucination OR calibration OR robustness OR interpretability OR generalization))	Topic; English; 1 Jan 2022-20 May 2026	22
arXiv	Preliminary Scoping	20 May 2026	(ti:("medication recommendation" OR "drug recommendation" OR "prescription prediction") OR abs:("medication recommendation" OR "drug recommendation" OR "prescription prediction"))	Title or Abstract; 1 Jan 2022-20 May 2026	189
arXiv	Formal Systematic	20 May 2026	(ti:("medication recommendation" OR "drug recommendation" OR "prescription prediction") OR abs:("medication recommendation" OR "drug recommendation" OR "prescription prediction")) AND (EHR OR "electronic health record" OR "clinical narratives" OR "multimodal fusion" OR "temporal dynamics" OR "drug-drug interactions" OR "knowledge graphs" OR ontologies OR hypergraphs OR Transformers OR GNNs OR "large language models" OR LLMs OR	Title or Abstract; 1 Jan 2022-20 May 2026	21

Source	Search stage	Final search date	Exact executed query	Limits applied	Records retrieved
			"retrieval-augmented generation" OR RAG OR hallucination OR calibration OR robustness OR interpretability OR generalization)		
			Total records retrieved from formal systematic searches (before deduplication)		132

S3. Targeted Venue-Specific Coverage Verification

The independent venue-specific screening was performed as a coverage check. All potentially relevant records identified through this process were already present in the 132-record source-based set, and no additional records were added to the master set.

Table S2. Independent venue-specific screening summary.

Venue	Type	Period covered	Actual screening method	Final screening date	Potentially relevant records identified	Additional records added to master set
KDD	Conference	1 Jan 2022-20 May 2026	Proceedings table-of-contents review with title/abstract keyword check	20 May 2026	8	0
WWW	Conference	1 Jan 2022-20 May 2026	Proceedings table-of-contents review with title/abstract keyword check	20 May 2026	6	0
RecSys	Conference	1 Jan 2022-20 May 2026	Proceedings table-of-contents review with title/abstract keyword check	20 May 2026	4	0
AAAI	Conference	1 Jan 2022-20 May 2026	Proceedings table-of-contents review with title/abstract keyword check	20 May 2026	12	0
IJCAI	Conference	1 Jan 2022-20 May 2026	Proceedings table-of-contents review with title/abstract keyword check	20 May 2026	7	0
CIKM	Conference	1 Jan 2022-20 May 2026	Proceedings table-of-contents review with title/abstract keyword check	20 May 2026	5	0
WSDM	Conference	1 Jan 2022-20 May 2026	Proceedings table-of-contents review with title/abstract keyword check	20 May 2026	3	0
ICDM	Conference	1 Jan 2022-20 May 2026	Proceedings table-of-contents review with title/abstract keyword check	20 May 2026	4	0
NeurIPS	Conference	1 Jan 2022-20 May 2026	Proceedings table-of-contents review with title/abstract keyword check	20 May 2026	9	0
ACM Multimedia	Conference	1 Jan 2022-20 May 2026	Proceedings table-of-contents review with title/abstract keyword check	20 May 2026	3	0
Knowledge-Based Systems	Journal	1 Jan 2022-20 May 2026	Journal website keyword search + issue-level ToC review	20 May 2026	11	0
Expert Systems with Applications	Journal	1 Jan 2022-20 May 2026	Journal website keyword search + issue-level ToC review	20 May 2026	9	0
Artificial Intelligence in Medicine	Journal	1 Jan 2022-20 May 2026	Journal website keyword search + issue-level ToC review	20 May 2026	7	0
IEEE TKDE	Journal	1 Jan 2022-20 May 2026	Journal website keyword search + issue-level ToC review	20 May 2026	6	0
Decision Support Systems	Journal	1 Jan 2022-20 May 2026	Journal website keyword search + issue-level ToC review	20 May 2026	5	0

Venue	Type	Period covered	Actual screening method	Final screening date	Potentially relevant records identified	Additional records added to master set
Journal of Biomedical Informatics	Journal	1 Jan 2022-20 May 2026	Journal website keyword search + issue-level ToC review	20 May 2026	8	0
			Total venue screening		107	0

Interpretation: The venue screening identified 107 potentially relevant records across the selected conferences and journals. After comparison with the 132-record master set, all were found to be already represented. This verification provided additional assurance regarding the coverage of the source-based search.

S4. Study Identification and Screening Summary

Table S3. Study identification and screening counts reported in Section 2.3 of the main manuscript.

Stage	Count
Total records retrieved from formal systematic searches (before deduplication)	132
Duplicate records removed	-39
Unique records for title/abstract screening	93
Records excluded at title/abstract screening	-37
Records for full-text assessment	56
Records excluded at full-text assessment	0
Studies included in qualitative synthesis	56

The exclusion breakdown for the 37 records is provided in Table 1 of the main manuscript.

Note: Foundational studies published before 2022 (e.g., GAMENet [51], SafeDrug [72], G-BERT [50], LEAP [79], Doctor AI [8], RETAIN [9], and MICRON [71]) are cited in the main manuscript for historical context and architectural foundations but are not included in the formal 56-study qualitative synthesis, which is restricted to the 2022-2026 publication window.

S5. Studies Included in the Qualitative Synthesis

Table note: The reference numbers in brackets correspond to the bibliography of the main manuscript. Preprints are indicated where the final venue had not been assigned at the time of search. Methodological categories are non-mutually exclusive and indicate the primary characteristics emphasized in this review.

Table S4. Studies included in the qualitative synthesis and their key characteristics.

No.	Reference	Year	Venue	Primary methodological characterization	Dataset(s) used
1	PharmaLLM [2]	2024	HCIS	LLM/RAG	Private
2	REFINE [4]	2023	NeurIPS	Graph	MIMIC-III
3	Chen et al. [6]	2025	KDD	Enhanced training	MIMIC-III + MIMIC-IV
4	Chen et al. [7]	2023	AAAI	Graph + safety	MIMIC-III
5	FLAME [12]	2025	NeurIPS	LLM/RAG	MIMIC-III
6	GPSRec [15]	2025	CIKM	Causal/debiased	MIMIC-III + MIMIC-IV
7	KRAM [18]	2026	ESWA	Robustness + knowledge	MIMIC-III
8	HI-DR [19]	2025	AAAI	Attention/Transformer	MIMIC-III + MIMIC-IV
9	VITA [20]	2024	AAAI	Attention/Transformer	MIMIC-III + MIMIC-IV
10	StratMed [21]	2023	arXiv	Robustness	MIMIC-III
11	CausalMed [22]	2024	CIKM	Causal/debiased	MIMIC-III
12	Trans-GAHNet [23]	2024	ESWA	Attention/Transformer + graph	MIMIC-IV
13	DGCL [24]	2023	JBI	Graph + enhanced training	MIMIC-III
14	MR-DTR [25]	2025	WWW	Temporal + causal	MIMIC-III + MIMIC-IV
15	CIDGMed [26]	2025	KBS	Causal/debiased + graph	MIMIC-III

No.	Reference	Year	Venue	Primary methodological characterization	Dataset(s) used
16	TreatRAG [27]	2025	RecSys	LLM/RAG	MIMIC-IV
17	EDRMM [28]	2025	BMC Bioinf.	Graph	MIMIC-III + MIMIC-IV
18	DNMDR [29]	2025	KBS	Graph + knowledge	MIMIC-III + MIMIC-IV
19	IMDR [30]	2025	KDD	Graph	MIMIC-III + MIMIC-IV
20	TEMPT [31]	2025	TOIS	Attention/Transformer + enhanced training	MIMIC-III + MIMIC-IV + eICU
21	LEADER [32]	2024	arXiv	LLM/RAG	MIMIC-IV
22	SHAPE [33]	2023	JBHI	Attention/Transformer	MIMIC-III
23	DKINet [34]	2023	arXiv	Knowledge integration	MIMIC-III + MIMIC-IV + eICU
24	MedAlign [35]	2025	ACM MM	LLM/RAG + multimodal	MIMIC-IV + MIMIC-CXR
25	PAUP [36]	2026	IP&M	Robustness + causal	MIMIC-III
26	MedGCN [37]	2022	JBHI	Knowledge integration + graph	MIMIC-III + Private
27	ACDNet [38]	2024	JBHI	Attention/Transformer	MIMIC-III
28	BiMoRec [39]	2025	ESWA	Graph	MIMIC-III
29	KindMed [40]	2023	arXiv	Knowledge integration	MIMIC-III
30	EGNet [41]	2025	Appl. Intell.	Graph	MIMIC-III
31	KATMed [45]	2026	JBHI	Attention/Transformer + safety	MIMIC-III + MIMIC-IV
32	RAG-CPMF [46]	2025	Pharm. Res.	LLM/RAG	Private
33	TAHDNet [54]	2022	JBHI	Attention/Transformer	MIMIC-III
34	CEHMR [55]	2023	AIM	Attention/Transformer + enhanced training	MIMIC-III
35	4SDrug [56]	2022	KDD	Safety-utility balancing	MIMIC-III
36	MetaCare++ [57]	2022	SIGIR	Robustness + causal	MIMIC-III + eICU
37	CAMeR [58]	2023	ICONIP	Causal/debiased	MIMIC-III
38	EXCERF [63]	2025	KBS	Knowledge + robustness	MIMIC-III + MIMIC-IV
39	ARMR [65]	2025	IJCAI	Attention/Transformer	MIMIC-III + MIMIC-IV
40	TAKECare [66]	2025	Inf. Fusion	Graph	MIMIC-III
41	PROMISE [67]	2024	IP&M	Multimodal + knowledge	MIMIC-III
42	COGNet [68]	2022	WWW	Causal/debiased + Transformer	MIMIC-III
43	SDRBT [73]	2025	JBHI	Safety-utility balancing	MIMIC-III
44	MoleRec [74]	2023	ACM MM	Graph + knowledge	MIMIC-III
45	AKA-SafeMed [75]	2024	Inf. Sci.	Recurrent + attention	MIMIC-III
46	SSPNet [76]	2025	IJCAI	Graph	MIMIC-III + MIMIC-IV
47	HKRec [77]	2024	TCSS	Recurrent	MIMIC-III
48	KEDRec-LM [78]	2025	arXiv	LLM/RAG + knowledge	MIMIC-IV
49	MedRec [80]	2022	TOIS	Graph + knowledge	MIMIC-III
50	KEHGCN [81]	2026	AAAI	Graph/hypergraph	MIMIC-III + MIMIC-IV + eICU
51	BH3-MedRec [82]	2026	TIST	Graph/hypergraph	MIMIC-III + MIMIC-IV + eICU
52	LAMO [83]	2025	arXiv	LLM/RAG	MIMIC-III + MIMIC-IV
53	RAREMed [84]	2024	SIGIR	Robustness + enhanced training	MIMIC-III + MIMIC-IV
54	MKAF-Net [85]	2025	Appl. Intell.	Multimodal	MIMIC-III
55	TCM-KLLaMA [87]	2025	CBM	LLM/RAG	Private (TCM)
56	DAI-Net [88]	2024	JBHI	Graph	MIMIC-III + MIMIC-IV + eICU

S6. Quality Appraisal of Representative Models

To complement the descriptive synthesis, the 51 models listed in Table 3 of the main manuscript were assessed using the seven reporting and evaluation criteria defined in Section 2.4. Each criterion was coded as adequate (2), partial (1), or inadequate/not reported (0). MetaCare++ [57] is retained in the descriptive model table because of its relevance to cold-start clinical modeling, but its cited study predicts diagnoses rather than medications; it is therefore marked not applicable and excluded from the aggregate appraisal. The aggregate results are based on 50 applicable models: Q1 49/1/0, Q2 43/7/0, Q3 46/4/0, Q4 47/3/0, Q5 36/6/8, Q6 19/25/6, and Q7 29/18/3 for adequate/partial/inadequate reporting, respectively. These scores are descriptive and should not be interpreted as a validated risk-of-bias scale or a ranking of model performance. Because Table 3 also includes foundational comparison models, this model-level appraisal does not redefine the formal 56-study PRISMA corpus reported in Table S4.

Criteria: Q1 task definition; Q2 dataset/preprocessing transparency; Q3 architecture/training reproducibility; Q4 MR-appropriate predictive evaluation; Q5 safety-aware evaluation; Q6 interpretability/evidence grounding; Q7 external validation/robustness/deployment.

Table S5. Model-level quality-appraisal matrix for the representative models in Table 3 of the main manuscript.

Model	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Total
AKA-SafeMed [75]	2	2	2	2	2	1	1	12
Trans-GAHNet [23]	2	2	2	2	2	1	1	12
VITA [20]	2	2	2	2	2	2	2	14
TAHDNet [54]	2	2	2	2	1	2	1	12
SHAPE [33]	2	2	2	2	2	1	1	12
KindMed [40]	2	2	2	2	2	1	2	13
MetaCare++ [57]	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
SafeDrug [72]	2	2	2	2	2	1	2	13
COGNet [68]	2	2	2	2	2	1	0	11
DGCL [24]	2	2	2	2	2	2	2	14
MKAF-Net [85]	2	2	2	2	1	1	2	12
PROMISE [67]	2	2	2	2	2	1	2	13
MedAlign [35]	2	2	2	2	2	1	2	13
MR-DTR [25]	2	2	2	2	2	1	2	13
SSPNet [76]	2	2	2	2	2	1	1	12
TreatRAG [27]	2	2	2	2	1	2	2	13
GPSRec [15]	2	2	2	2	2	1	2	13
DKINet [34]	2	2	2	2	2	2	2	14
RAREMed [84]	2	2	2	2	2	0	2	12
StratMed [21]	2	2	2	2	2	2	2	14
G-BERT [50]	2	2	2	2	0	1	0	9

Table S5 (continued). Model-level quality-appraisal matrix.

Model	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Total
LEADER [32]	2	2	2	2	0	0	1	9
ARMR [65]	2	2	2	2	0	1	1	10
KEDRec-LM [78]	2	2	1	1	0	1	1	8
ConfidenceCalib [6]	2	1	2	2	1	0	2	10
MedRec [80]	2	2	2	2	0	2	2	12
MedGCN [37]	2	2	2	2	0	1	2	11
CEHMR [55]	2	2	2	2	2	2	1	13
4SDrug [56]	2	2	2	2	2	1	2	13
CausalMed [22]	2	2	2	2	2	2	1	13
CAMeR [58]	2	1	1	2	2	2	2	12
KEHGCN [81]	2	2	2	2	2	2	2	14
BH3-MedRec [82]	2	2	2	2	2	2	2	14
DNMDR [29]	2	2	2	2	2	2	1	13
ACDNet [38]	2	2	2	2	2	1	1	12
RAG-CPMF [46]	2	2	1	2	2	2	1	12
TCM-KLLaMA [87]	2	1	2	2	1	1	1	10
MoleRec [74]	2	1	2	2	2	2	1	12
DAI-Net [88]	2	1	2	2	2	2	2	13
EDRMM [28]	2	2	2	2	2	1	1	12
EGNet [41]	2	2	2	2	2	2	2	14
CIDGMed [26]	2	2	2	2	2	2	1	13
TEMPT [31]	2	2	2	2	0	0	2	10
PharmaLLM [2]	1	2	2	1	1	0	1	8
FLAME [12]	2	2	2	2	2	1	2	13
GAMENet [51]	2	2	2	2	2	1	0	11
TAKECare [66]	2	2	2	2	2	1	2	13
EXCERF [63]	2	2	2	2	2	2	2	14
KRAM [18]	2	1	1	1	0	0	2	7
KATMed [45]	2	2	2	2	2	1	2	13
PAUP [36]	2	1	2	2	2	1	2	12

Scoring: 2 = adequately reported; 1 = partially reported; 0 = inadequate or not reported; N/A = MR-specific rubric not applicable. **Interpretation:** the appraisal primarily reflects reporting and evaluation completeness rather than comparative model quality. The clearest gap is interpretability/evidence grounding, where only 19 of 50 applicable models received an adequate rating; safety evaluation and evidence for generalization were also less consistently reported than task definition and predictive evaluation.